<u>Data Management Technologies</u>
<u>Unit 6 – Consolidation Task</u>

Activity 1: Understanding Concepts

Normalisation is a process by which a database is structured to reduce data dependency and eliminate redundancies. Normalised data is spread across multiple tables, so that fields do not depend on each other as much. In a virtual cookbook compilation, the data objects include ingredients, recipes and the cookbooks the recipes were sourced from.

RecipeID	RecipeName	Ingredients	CookbookID	CookbookName
01	Pancakes	flour, butter, milk	1	Swift Breakfasts
02	Jacket Potatoes	potatoes, butter, salt	2	Hot Lunches
03	Omelette	eggs, milk, salt	1	Swift Breakfasts

The data is unnormalised, as there are multiple ingredients in each cell, as well as partial dependencies (cookbook name depends only on cookbook ID) and transitive dependencies (ingredients depends on recipe name, which depends on recipe ID).

With 1NF normalisation, each ingredient is represented on its own row, so there are not multiple values in a cell. As such, the first column is transformed as below:

RecipeID	RecipeName	Ingredient
01	Pancake	flour
01	Pancake	butter
01	Pancake	milk
02	Jacket Potatoes	potatoes
02	Jacket Potatoes	butter
02	Jacket Potatoes	salt
03	Omelette	eggs
03	Omelette	milk
03	Omelette	salt

With 2NF normalisation, partial dependency between RecipeID and Ingredient is removed. Ingredients becomes a new table:

IngredientID	Ingredient
001	flour
002	butter
003	milk
004	potatoes
005	salt
006	eggs

Ingredients can be linked to the recipe table as below:

RecipeID	**IngredientID**
01	001
01	002
01	003
02	004
02	002
02	005
03	006
03	003
03	005

Finally, to achieve 3NF, the transitive dependency between Cookbook Name and Recipe ID should be removed. (Cookbook Name depends on Cookbook ID, which depends on Recipe ID). Cookbook Name should be moved to a new table, containing Cookbook Name and Cookbook ID.

CookbookID	**CookbookName**
1	Swift Breakfasts
2	Hot Lunches

Activity 2: Identifying Normalisation Stages

The example given is a relational database structure for a Fixed Asset Register for a supermarket.

AssetID	**AssetName**	**LocationName**	**PurchseTransactionID**	**PurchaseCost**	**PurchaseDate**
001	Shelving	Shopfloor, Warehouse	2021/01	£3000	12/06/2021
002	Chest Freezer	Shopfloor, Warehouse	2020/93	£1000	30/02/2020
003	Conveyor belt	Checkout	2021/02	£2000	10/01/2021
004	Desk	Admin	2014/17	£500	14/08/2014

Repeating groups (1NF issues) - Location Name contains multiple values in some cells.
Partial dependencies (2NF issues) - There are partial dependencies. Location name depends only on Asset ID, and not Asset Name or the other values.
Transitive dependencies (3NF issues) - The fields purchase cost and purchase date depend on purchase transaction ID, which depends on Asset ID.

Activity 3: Table Conversion Practice

Convert the following unnormalised dataset into tables satisfying 1NF, 2NF, and 3NF:

Student	**Course**	**Instructor**	**Instructor Office**
Alice	Math	Dr. Smith	A201
Alice	Science	Dr. Jones	A202
Bob	Math	Dr. Smith	A201
Bob	Science	Dr. Jones	A202

First Normal Form

The original table satisfies 1NF, as the data is contained in atomic values, which each cell containing only one value.

Second Normal Form

In the original table, Instructor depends only on Course, which is a partial dependency, as it depends on only part of the composite key (student and course)

The table can be split as below:

Student Enrolments	
Student	**Course**
Alice	Math
Alice	Science
Bob	Math
Bob	Science

Course Instructors		
Course	**Instructor**	**Office**
Math	Dr. Smith	A201
Science	Dr. Jones	A202

Third Normal Form

The original table did not satisfy 3NF, as there was transitive dependency. Each instructor's office depends on instructor, which in turn depends on the primary keys (student and course). This is remedied below, by moving Office to a new table:

Student Enrolments		
Student	**Course**	**Instructor**
Alice	Math	Dr. Smith
Alice	Science	Dr. Jones
Bob	Math	Dr. Smith
Bob	Science	Dr. Jones

Instructor Assignments	
Course	**Instructor**
Math	Dr. Smith
Science	Dr. Jones

Instructor Offices	
Instructor	**Office**
Dr. Smith	A201
Dr. Jones	A202

Activity 4: Higher Normal Forms Case Studies
Examine provided datasets with 4NF, 5NF, and 6NF issues. Decompose them into appropriate tables.

<u>Fourth Normal Form</u>

Additional rows have been added to the table below to illustrate the fourth normal form.

Student Enrolments		
Student	**Course**	**Module**
Alice	Math	Calculus
Alice	Math	Algebra
Alice	Science	Biology
Alice	Science	Physics
Bob	Math	Calculus
Bob	Math	Algebra
Bob	Science	Biology
Bob	Science	Physics

Within the table, there are independent repeating relationships in the module column, which violate 4NF. This can be fixed by splitting the table into two tables.

Student Enrolments	
Student	**Course**
Alice	Math
Bob	Science

Course Modules	
Course	**Module**
Math	Calculus
Math	Algebra
Science	Biology
Science	Physics

Fifth Normal Form

The fifth normal form eliminates tables that can be obtained by joining smaller tables. If a table can be split and rejoined without losing or adding wrong data, then it violates 5NF.

To illustrate 5NF, we can complicate the table by adding more data so that more than one teacher teaches a course.

Course Enrolments		
Teacher	**Student**	**Course**
Dr. Smith	Alice	Maths
Mr. Alex	Charlie	Maths
Dr. Jones	Alice	Science
Mrs. Bolton	Charlie	Science

The table shows three relationships: which teacher teaches which subject to which student. However, it can be split into three two-way tables without losing or adding wrong data.

The table can be remedied as below. The information is split into three tables, each with two columns. No data is lost and the original table can be obtained by joining the split tables.

Teacher Registrations	
Teacher	**Course**
Dr. Smith	Maths
Mr. Alex	Maths
Dr. Jones	Science
Mrs. Bolton	Science

Student Registrations	
Teacher	**Student**
Dr. Smith	Alice
Mr. Alex	Charlie
Dr. Jones	Alice
Mrs. Bolton	Charlie

Course Enrolments	
Course	**Student**
Maths	Alice
Science	Alice
Maths	Charlie
Science	Charlie

Sixth Normal Form

A table is in 6NF when it cannot be decomposed further, and every table has only the primary key (even a composite primary key), and one attribute.

6NF is useful in temporal databases. In temporal databases the data changes with time, and so by separating tables, only the relevant table must be changed when a value changes.

This is illustrated below using the college scenario. This table (pre-transformation) records month-end textbook inventory quantities and prices. There is a composite primary key: **PK = (Book, Date)**

Maths Textbook Inventory			
Book	**Date**	**Price**	**Quantity**
Functional Skills in Mathematics by J.H. Smith	31/01/2026	£20	100
Functional Skills in Mathematics by J.H. Smith	28/02/2026	£20	80
Functional Skills in Mathematics by J.H. Smith	31/03/2026	£25	80

Storing data as above makes it easy to calculate month-end inventory values (it's simply quantity multiplied by price). However, whenever either inventory or price changes, a new row must be added, even though only one of the values changed that month.

This can be avoided through the 6NF, where the data is stored in two separate tables. Now, a new row is only added when a value in the table changed during the month. This reduces the number of figures to be updated.

The transformed tables are:

Maths Textbook Prices		
Book	**Date**	**Price**
Functional Skills in Mathematics by J.H. Smith	31/01/2026	£20
Functional Skills in Mathematics by J.H. Smith	31/03/2026	£25

Maths Textbook Quantities		
Book	**Date**	**Quantity**
Functional Skills in Mathematics by J.H. Smith	31/01/2026	100
Functional Skills in Mathematics by J.H. Smith	28/02/2026	80

However, the 6NF takes much space, as it increases the number of tables in the database. Further, as noted earlier, in a business or accounting context, the table would have been more useful before transformation, as it facilitated the calculation on monthly inventory values. For reasons like these, 6NF is not used often in practice.

Sources:

- Connolly, T., & Begg, C. (2015). *Database systems: A practical approach to design, implementation, and management* (6th ed.). Pearson.

- W3schools. (2026). *Database normalization*. https://www.w3schools.in/dbms/database-normalization#fifth-normal-form